# Choose a motion method

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Choose a motion relationship. Explain what must be true for you to use it.

**SUCCESS CRITERIA - I can...**

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “constant” and “interval” to explain your answer. Sentence starters are optional.

**constant**: not changing during the stated interval

**interval**: the time or range between two stated values

**acceleration**: rate of change of velocity

## 01 Warm-up: bring back an earlier idea

A cart starts at **3 m** and moves right at **5 m/s** for **2 s**. **Predict final position**. Allowed endpoints are **11** to **15 m**; **state this interval** and one thing you are treating as true.

**Word help:** final = at the end of the interval; subscript f

**Sentence starter:** I predict … . I used … to make this prediction because … .

**Marzano check - circle:** ① ② ③

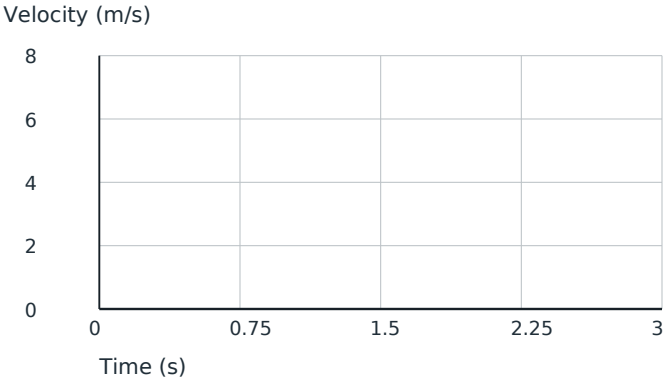
## 02 Worked example: explain the prediction

**Reference:** $v_f = v_i + a\,\Delta t$; $\Delta x = \frac{1}{2}(v_i + v_f)\Delta t$.
Constant acceleration; i = initial, f = final. At constant velocity, $\Delta x = v\,\Delta t$.

**Worked reasoning:** From rest, a cart accelerates uniformly at **2 m/s²** for **3 s**. Final velocity=**6 m/s**; average velocity=(**0**+**6**)/**2**=**3 m/s**; displacement=**3**×**3**=**9 m**.

**Your turn:** Complete the cart's velocity-time graph. **Shade the triangular area under** the line and use it to **check the 9 m** displacement.

**Sentence starter:** In my work, I used … to show … . I labeled it … .

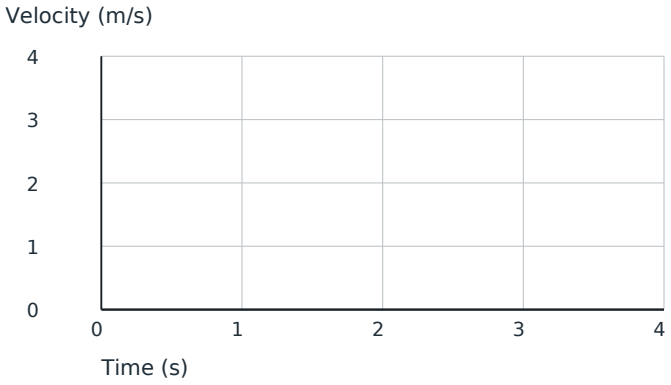



**Marzano check - circle:** ① ② ③

## 03 Choose the relationship

A robot drives straight at a steady **3 m/s** for **4 s**.
1. **Sketch its velocity-time graph**.
2. **Decide whether** a constant-velocity or acceleration relationship fits, then **find its displacement**.

**Sentence starter:** I found … [unit]. This value means … in this situation.




**Marzano check - circle:** ① ② ③

## 04 Build final velocity

A remote-control car starts at **2 m/s** *east* and gains **2 m/s** each second for **3 s**. **Complete the table**. **How much** velocity does it gain altogether, and what is its final velocity?

**Sentence starter:** I notice … . This supports … because … .

| Time (s) | Velocity east (m/s) |
|---|---|
| 0 | 2 |
| 1 | 4 |
| 2 | 6 |
| 3 | ? |

**Marzano check - circle:** ① ② ③

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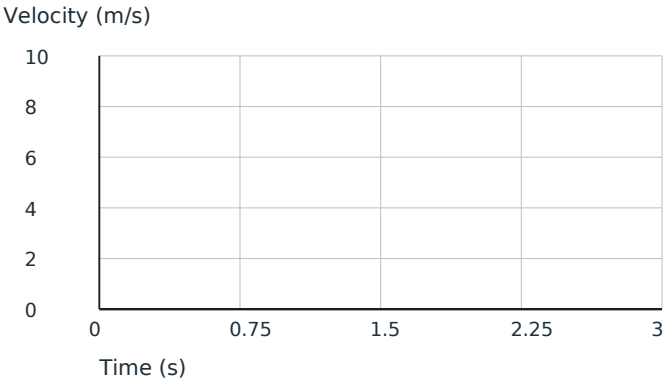
Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Find the distance while speeding up

Use the car's velocities from question **4** to **complete the graph**. Its acceleration is constant. **Find its average velocity** and displacement over the **3 s**. **Use the graph**'s area to check.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

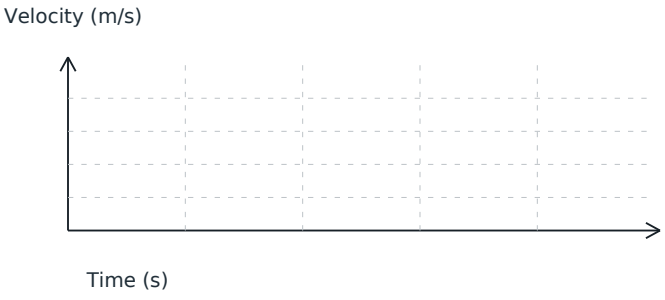



**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A cart travels *west* at **10 m/s** and *west* at **3 m/s 2 s** later. Is it speeding up? **Find acceleration size** and direction. **Sketch the velocity**–time graph; **label the given values**.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

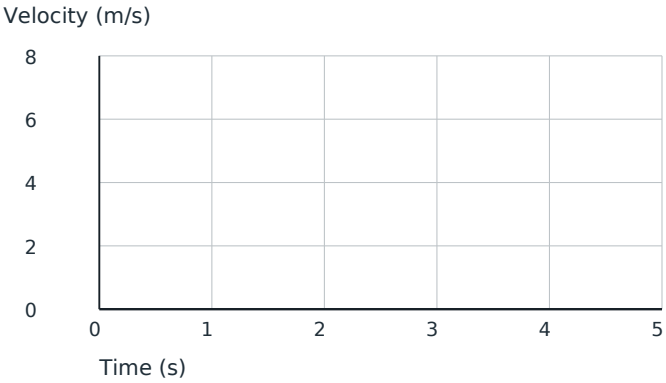



**Marzano check - circle:** ① ② ③

## 07 Exit check

In a launch test, a small rover speeds up steadily from rest to **6 m/s** *east* in **5 s**. Plot the two endpoints and connect them. **Find acceleration**, average velocity and displacement.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

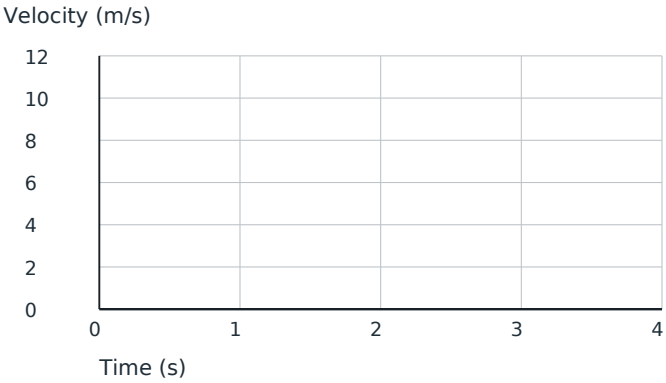



**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the prediction

1. Create two different motions that travel **8 m** in **4 s**: one at steady speed and one speeding up uniformly from rest.
2. **Draw their velocity-time graphs**.

**Sentence starter:** Both ... . They differ in ... because ... .




**Marzano check - circle:** ① ② ③